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To cite this article: Marios Trigkas, Christos Anastopoulos, Ioannis Papadopoulos & Dimitra Lazaridou (2020) Business model for developing strategies of forest cooperatives. Evidence from an emerging business environment in Greece, *Journal of Sustainable Forestry*, 39:3, 259-282, DOI: [10.1080/10549811.2019.1635031](https://doi.org/10.1080/10549811.2019.1635031)

To link to this article: <https://doi.org/10.1080/10549811.2019.1635031>



Published online: 26 Jun 2019.



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Business model for developing strategies of forest cooperatives. Evidence from an emerging business environment in Greece

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ABSTRACT

Forest cooperatives are essentially a form of entrepreneurship. However, they are confronted by various problems and they cannot pay enough attention to value creation. The aim of this paper is twofold: (a) the identification of the structural factors supporting the function of Greek forest cooperatives and the exploration of the possibilities of applying business model innovation to them; (b) the investigation of how the development of an integrated business model can help cooperatives stay ahead in the innovation game, while securing sustainability. By applying business modeling procedures to the forest cooperatives of the Region of Thessaly (Greece) this paper presents, the implementation of innovative development strategy, within the current legislative framework. Results show that the business model approach gives them the ability to develop a shared vision for value creation and delivering through the establishment of a framework related to sustainable forest management based on creation of bonds for achieving a win-win strategy, satisfying the increasing needs for innovative forest-based products, which constitutes a challenge for the sustainable management of forests. We argue that this is the first time this approach has been made regarding forest cooperatives in Greece in relation to the emerging parameters of the general business environment.

KEYWORDS

Business model; business model canvas; forest cooperatives; business strategy; business model innovation; forestry

Introduction

Current needs for every organization, such as economic development, diversification and enhancement of competitiveness, require the adoption of innovations as well as the introduction of new products, services and knowledge, aiming at creating value and meeting the modern citizens' needs (Alstete, 2007; Altman & Iles, 1998; Bierly, Kessler, & Christensen, 2000; Nonaka, Toyama, & Nagata, 2000). Especially during times of economic recession, such as the deep one that the Greek economy is experiencing, Drucker (1985) cites that, *“adaptable organizations are those that will survive”*, as well as that *“challenge for the development is not the direction an organization will set, but what it will be able to change in order to succeed it”*. This view may be the most important asset for an organization in its endeavor to survive. It is very important for an organization to be able to perceive the modern needs of society and to reorganize its structures and culture in such a way that it can fulfill them. Additionally, by understanding the current needs of society, it enables every organization to restructure its identity and culture in order to meet them.

Forest cooperatives, like all joint ventures, are essentially a form of entrepreneurship. They undertake an activity aiming at the economic benefit. The main difference between a forest cooperative and any other form of entrepreneurship is that a cooperative aims to improve its services in order to increase the income of its members, while the latter aims at profit maximization for its owners. Focusing more on forest workers' cooperatives, their purpose is to care for, exploit, regenerate and protect a specific forestland, taking into consideration the current regulations of the forestry legislation (Cace, Arpinte, Cace, & Cojocar, 2011). In general, typical services offered by a forestry cooperative include: (i) forest product marketing, (ii) management to improve forest productivity and profitability and (iii) enhancement of social relationships (Cobia, 1989). Further benefits of participating in a forest joint venture include improved services, trusted information, economies of scale, reduced costs, and better access to markets (Hull & Ashton, 2008). In addition, Krott (2005, p. 96) points out that the collaboration and unity of the forest sector are emphasized by the cooperatives as being their goal.

However, forestry cooperatives thrive in Europe, Japan and Australia (Kittredge, 2005), and the existing ones are faced with numerous, various problems. Among them, the literature highlights the fact that they seldom operate as well-functioning organizations. This is due to suffering by structural problems such as, inadequate management, limited access to funding resources and skilled labor along with lack of legislative support, they cannot pay enough attention to research and value creation, and they are unable to act efficiently in order to address forestry and environmental problems (Atmiş, Günşen, Lise, & Lise, 2009; Coulibaly-Lingani, Savadogo, Tigabu, & Oden, 2011; Hull & Ashton, 2008). As it arises, forestry joint ventures operate in a competitive business environment. Thus, it is necessary for forest cooperatives to look at the forest management from a strategic perspective, taking under consideration the multifunctional character of forestry and its contribution to sustainable development. However, one of the major problems currently facing the forestry sector is the low profitability of forest production (Melo, Cunha, & Ferreira, 2017). This implies a greater professionalization of forest management, incorporating more scientific knowledge and technical-professional skills and the adoption of new business models (Melo et al., 2017). Cooperatives' lucrative activity in the future requires diversification strategy and development of innovative activity (Kajanus et al., 2019; Trigkas, Andreopoulou, Papadopoulos, & Kitsouli, 2016; Trigkas, Papadopoulos, & Karagouni, 2012).

Even for the application of new economic models, such as this of circular bioeconomy in the forestry sector which implies a more efficient resource management, research highlights that the business models are dominated by traditional practices (D'Amato, Veijonaho, & Toppinen, 2018). New approaches are required for researchers and policy-makers to advance the sustainability transition of forestry in this vein as well.

Therefore, the aim of this paper is twofold: (a) the identification of the structural factors supporting the function of Greek forest cooperatives and the exploration of the possibilities of applying business model innovation to them; (b) the investigation of how the development of an integrated business model can help forest cooperatives stay ahead in the product innovation game, forming a concrete strategy, while improving the quality of Greek forests and environmental protection contributing to sustainable and multifunctional forestry. By applying business modeling procedures to the forest cooperatives of the Region of Thessaly (Greece), this paper presents, for the first time in Greece, the implementation of innovative development strategy within the current legislative

framework for the operation of forest cooperatives (Law 4423/2016: Forest Cooperative Organizations and other provisions, FEK 182/A/27-9-2016). The results of this study may provide some guidelines to policymakers on how to harness economic and environmental potentials to boost the forest-based economy and secure sustainable forestry and its multifunctional role.

Forest management and cooperatives in Greece

Existing situation

Greece is a mountainous country, with extended forested areas, over 51% of the total land area, which are shaped by several species (Kyriakopoulos, Kolovos, & Chalikias, 2010; Ministry of Rural Development and Food, 2006). The shrub or herbaceous vegetation covers 4,182,000 ha (31.7%) and almost 213,379 ha (1.6%) are areas with sparse or no vegetation. Regarding the proprietary status, the dominant form of forest ownership is public forests, at 74.1%. A percentage of 9% is community-owned forests, while 6.5% of forests are private and 10.4% continue to be owned by others (social welfare organizations, local authorities, and others).

Forest ecosystems are very valuable to every region, not only for their environmental and social functions but also for the economic wealth they generate. The contribution of the forestry sector represented less than 1.32% of the total Gross Domestic Product (GDP) produced in the country. This most likely resulted from the fact that Greek forests are of a protected nature, while the benefits arising from this function are neither valued in monetary terms, nor recorded in the national accounts. Moreover, the investments that take place in Greek forestry primary as well secondary production still remain extremely limited (Kyriakopoulos et al., 2010). In addition, the productivity level of Greek forests is low, due to several organizational, management and cultural conditions that exist in Greek rural areas in general and in forestry specifically such as, limited access to finance and skilled labor, poor infrastructure, small size of the firms, etc. (Karagouni, Trigkas, Papadopoulos, & Mpasdekis, 2017). At the same time, changing policies such as reductions in forestry support increases pressures on forest business especially when combined with changing market trends. (Karagouni et al., 2017). This is a reason that forest management cannot always achieve its goals for the improvement and upgrading of Greek forests (Chalikias, Kyriakopoulos, & Kolovos, 2010).

The basic tool for the management and exploitation of Greek forests is forest management studies, whose duration is usually 10 years. Based on these studies and in accordance with the legislation in force, the public forest service leases the production to forest workers' cooperatives, which pay a lease depending on the volume of the timber expected to be produced. Cooperatives are responsible for wood harvesting with the obligation of attributing part of their revenues as tax to the Greek State (10%) and a lower percentage of the profits to the municipality that owns the forest they exploit (5%).

Regarding forest cooperatives, official data shows an increase in numbers, but a steady decline in the number of forest workers in Greece. In the year 2004, there were 600 forest cooperatives, which numbered about 20,000 members. In 2008 cooperatives increased to 810 with about 8,450 registered members, while workers' numbers shrank further in 2011 (7,470 members) (Ministry of Environment, Energy and Climate Change, 2014). As it is

currently estimated, the number of forest cooperatives is around 1,000, as opposed to the number of active workers that is constantly decreasing. In Greece, there is an increasing need for the forest workers' cooperatives to be restructured, in order to effectively respond to the challenges of the competitive environment in which they operate, let alone taking into account both the large and untapped production capacity of forest ecosystems and the fact that the private sector has some difficulties in growing in the forest sector, due to the institutional commitments and the proposed environmental policy and strategy at national and European level (European Commission, 2013, 2015, 2017; Hansen, 2016). Further reasons that make reorganizing of cooperatives imperative are: (i) their difficulty to acquire enough financial resources to accomplish their missions (ii) the weakness of effective management of forest ecosystems by the Greek public forestry agencies, mainly due to lack of sufficient staff, (iii) their difficulty to create any added value to the wood sold, (iv) their persistence on wood as the main product, (v) the low profit margin from selling of products (due to their disposal only to traders) (vi) the lack of personnel who want to gain further training, (vii) the short duration of employment in forest sector. Another oddity prevalent in Greece is the fact that in some areas the workforce is not enough to accommodate forest harvesting needs, whereas in other regions a large number of forest cooperatives turns forest harvesting into a problem (Tsioras, 2010).

So, by overcoming the aforementioned problems and function in a modern and constantly evolving context, and by applying new practices and technologies many social, economic and environmental benefits will be generated, especially nowadays where the demand for firewood has risen (Arabatzis & Malesios, 2013). In addition, the empowerment of Greek forest cooperatives will contribute to the further protection of the environment, the improvement of unfavorable living conditions of the population of less rural areas, the increase of forest workers' employment, as well as strengthening of social responsibility for forest ecosystems.

The new legislation framework for forest cooperatives in Greece

In the past, forest cooperatives were under the same legislative framework as agricultural cooperatives. This fact had several negative features, including the prioritization of issues related to agricultural cooperatives, which was overwhelming, disregarding forestry issues and neglecting the particularities of the forestry sector. A significant change in the function of forest cooperatives came with the law 2810/2000. This law reduced the number of registered members required to set up a cooperative to just seven and made the conditions for registration in a forestry cooperative more flexible. The amendments made by the law had as a consequence the fragmentation of forest cooperatives. It had been acknowledged that the application of the above law would lead to the creation of cooperatives with the minimum number of members, with the purpose to be able to claim the assignment of forest compartment or stand from the local Forest Services.

A new law on forest workers' cooperatives was designed to eliminate deficiencies in the previous legislation. The law 4423/2016 (Forest Cooperative Organizations and other provisions, FEK 182/A/27-9-2016) introduces certain changes in the existing type and gives further perspective to the Greek forest cooperative. The new legislation provided, inter alia, an increase in the number of members required to establish a forest cooperative. It requires at least 21 workers to establish a cooperative. What is more, forest workers

need to be suitably trained. Their skills have to be certified, and for those who are going to be occupied for the first time, the provision of related training is obligatory. Most importantly, the 4423/2016 law granted more powers to the forest cooperatives, given that apart from forestry operations, they are also active in the handling, marketing, and processing of forest products. It introduces the mandatory disposal of forest products by auction, excluding any direct agreement with the traders. Additionally, consultancy, supply of means and resources, marketing, trading of products, training, investments in wood processing industries, etc., of forest cooperatives can be done by a Forest Cooperatives Association, which can be established by five or more cooperatives as members, on condition that they belong to the same prefecture. The law also sets a more organized and supervised operating framework, including the financing of forest cooperatives, the provision of tax incentives, but also the obligation to have an internal operation regulation and to be supervised by the Forest Directorates of the prefecture where they belong. The above-mentioned provisions of the new law aim at creating robust and competitive cooperatives that ensure not only income for workers, but also strengthen social responsibility for the protection of the forest environment.

Business modeling

Conceptual framework

Entrepreneurship is a vague meaning that several scholars approach through the business model's theory to analyze its structural elements. Business models describe the way of "doing business" for an organization (Chesbrough & Rosenbloom, 2002; Magretta, 2002). According to another approach, a business model describes how an economic actor chooses to operate and interact with several markets in order to create and deliver value (Chesbrough & Rosenbloom, 2002; Zott & Amit, 2008). It is an organized set of structural elements encompassing the flows of costs, value proposals, revenues, and profits (Teece, 2018). Thus, a business model describes the way how a firm or an organization creates, captures and delivers value in several contexts, while fulfilling its goals. Business model is referred to as the outline of how an enterprise operates; namely, it describes the rationale of how a business chooses its customers, how it differentiates its products and services, how it acquires and disposes particular resources, and how it understands the specific customer needs (Osterwalder & Pigneur, 2010). Under this perspective, a business model is as a description of the activities and the relations developed among them, in order for a firm to implement its strategy (Arend, 2013). Zott and Amit (2010) define business models as a system of correlating activities in the firm that goes beyond the direct firm's environment and expands its boundaries. In a similar tone, Chesbrough (2007) describes a business model as a group of activities, from raw materials acquisition to the satisfaction of the needs of the final consumer. This approach is also adopted by several other scholars (DaSilva & Trkman, 2014; Massa & Tucci, 2013). They see business models as an integrated system understanding of how a firm organizes its activities for value creation. This point of view is aligned with Porter's approach of the firm as a chain of different activities for value creation (Porter, 1985).

A different approach is supported by authors who propose describing business models as a group of crucial structural elements as important parts of businesses' operation. Several relevant researches suggest this approach. Johnson, Christensen, and Kagermann

(2008) propose a structure based on the customer value proposition, profits creation and acquisition by the firm, key resources, and processes. Osterwalder and Pigneur (2010) propose a group of nine structural factors: key partners, key activities, key resources, value proposition, customer relationships, channels, customer segments, cost structure, and revenue streams, known as the Business Model Canvas. It is a model management strategy for developing new or recording existing business models. The Business Model Canvas (BMC) is a schematic representation for the interaction of different elements to capture and deliver value to customers (Castillo-López, Guirao, Roldán-Reyes, Macias, & Duque, 2019). It helps businesses align their activities capturing the possible compromises. Osterwalder's (2004, p. 2010) business model canvas is quite widespread in the entrepreneurial context because it helps entrepreneurs to make sense of "doing business" (Blank, 2013; Massa & Tucci, 2013; Trimi & Berbegal-Mirabent, 2012). This can be interpreted under the cognitive science approach of entrepreneurship as well. In contrast with the typical economic approach, the use of the organizational approach of the firm, tells us that the use of tools such as cognitive maps, can help reveal differences in the character of entrepreneurs and managers (Malmström, Johansson, & Wincent, 2015).

Business modeling and strategy

Business strategy research has efficiently analyzed how the use and the nature of strategic variables such as firm resources, capabilities and activities, and the environment in which the firm operates affect its performance. However, business strategy deals with making decisions that incorporate the sum of business activities in the pursuit of company goals. According to Cosenz and Noto (2018), business models offer an opportunity to bridge the gap between our knowledge about the effects of individual variables on firm performance and the need to understand how the relations among them play role and guide performance. The existence of a concrete strategy based on coherent and aligned elements secures the homogeneity of decision-making and supports an organization's business model (Rumelt, 2011). A strategy can be defined as "*a coherent set of analyses, concepts, policies, arguments, and actions that respond to a high-stakes challenge*" (Rumelt, 2011). Strategic analysis leads to the selection of a particular business model for the firm to create and deliver value, by gaining simultaneously a strategic competitive advantage in global markets. Many times, this leads businesses to abandon their old business model for a new one, in order to differentiate in international markets (Casadesus-Masanell & Ricart, 2011). Marketing strategy also plays a substantial role in designing a viable business model. Successfully designed business models will be able to expand across multiple market segments but knowing which of them constitute the market – target is extremely important. This is in line with the fact that marketing strategy plays a crucial role in the successful development and implementation of a business model (Teece, 2017).

Furthermore, international policies and society's needs, have led to the development of a new stream of entrepreneurship which pursues social and/or ecological goals, along with the classical commercial and profit orientation (Battilana & Lee, 2014). This has been supported by contemporary research highlighting entrepreneurship to address the social and environmental problems through providing new ideas for large-scale change in the economic and business model that prevails in mountainous and rural areas, ensuring the participation of local society in entrepreneurship known as sustainable entrepreneurship

or social enterprises (Trigkas, Papadopoulos, Karagouni, & Lazaridou, 2017). A very recent study by Todeschini, Cortimiglia, Callegaro-de-Menezes, and Ghezzi (2017) describes this evolution as putting sustainability and innovation in business models, which follow shared values and challenges for global societies and economies. Such entrepreneurs and their ventures are currently believed to be of significant economic, institutional, and academic interest (Haigh & Hoffman, 2012).

Business modeling in forestry and forest cooperatives

Business modeling is a vague topic that has been extensively analyzed in the business theory context (Ausrød, Sinha, & Widding, 2017; Chesbrough & Rosenbloom, 2002; Fjeldstad & Snow, 2018; Hoveskog, Halila, Mattsson, Upward, & Karlsson, 2018; Ibarra, Ganzarain, & Igartua, 2018; Osterwalder & Pigneur, 2010; Ritter & Lettl, 2018; Szulecka, Obidzinski, & Dermawan, 2016; Toro-Jarrín, Ponce-Jaramillo, & Güemes-Castorena, 2016). Nevertheless, our extended literature review revealed that there is extremely limited research relative to the issue, namely applying business modeling methods in forestry and forest cooperatives. Some related contemporary research on the issue is following.

Most contemporary research, like this of Kajanus et al. (2019), which has analyzed existing business models in the forestry sector of eight different European countries by applying the BMC approach, has highlighted that the changing operational environment forces the traditional forestry industry to adapt to emerging environments adopting several innovations to meet consumers' needs. According to this representative research of the forestry sector in northern and middle Europe, the business models are mostly grounded on traditional forestry and mainly include either new services or organizational improvements to meet customer needs by controlling the required transaction costs.

The work of Aguilar-Støen (2018) has to do with the evaluation of new legislation for social forestry movements and science-policy networks in Guatemala. The research recognizes the importance of property rights to land with strong social, economic and institutional impact, other than private property. Findings also indicate that community organizations can, by networking and developing synergies with institutional and scientific organizations, participate in decision-making and by that, in introducing a "bottom-up" environmental governance, while other research (Szulecka et al., 2016) points to socially oriented community forest management as highly praised by its stakeholders, being able to improve rural livelihoods and secure environmental benefits.

Another research of Atmiş et al. (2009), regarding forest cooperatives in Turkey, indicates that active members, sufficiency of work, use of published material, and the liquidity of the cooperative are important variables to explain variations in the decision of cooperatives to operate in forestry. According to the same study, there is a significant correlation between wood production and levels of participation in a forest cooperative, indicating that a productive forest is strongly related with high percentages of forest cooperatives' participation in forestry.

García-López (2018) argues a possible reconsideration of dynamic relations between foresters and local communities in Mexico, not only as dominant or unequal but also as symbiotic. He supports a win-win situation formed by mutual gains and co-dependency for jobs creation, continuing of forest activities and social wellbeing.

A very interesting research is this of Vega and Keenan (2016), who applied New Institutional Economic theory to identify the key organizational features and potential inherent weaknesses of Community Forestry Enterprises (CFEs). They argue that, enterprises where ownership rights are not belonging only to the investor, they face important management and collective decision-making challenges. Public support has played an important role in building the necessary supportive mechanisms for the development of these enterprises. Similar are the results of Ludvig, Wilding, Thorogood, and Weiss (2018), concluding that continuous cooperation and networking are necessities for collective engagement in forestry.

Research design and methods

Measures and data collection

The survey was conducted among 108 forest cooperatives operating in the region of Thessaly, Greece. In this region there exist a sufficient number of forest cooperatives, so it was treated as a pilot case which could be extended geographically. The survey was implemented from October 2017 to March 2018. All participating cooperatives are registered in the National Register on active cooperatives of Greek Ministry of Environment and Energy. For the purpose of the survey, a well-structured questionnaire was developed, which was carried out with the heads of the cooperatives. The questionnaires were applied, mainly, in face-to-face interviews. Where the completion of the questionnaires through personal interview was not possible, data were collected using e-mail exchanges and participants were given key information. The study regards the registration of the responses from all the forest cooperatives' population in the region of Thessaly. The survey generated 65 useable observations in total, which gave a response rate of 60.2%, and it is considered as absolutely satisfactory.

The development of the main research methodology and this of the questionnaire was based on building a tailor-made business model canvas for the forest cooperatives in the region, taking under consideration the specific characteristics of their operation as well as the context that the new legislation framework sets for them. We used the Osterwalder and Pigneur's (2010) Business Model Canvas approach. As mentioned, they propose a group of nine structural factors: key partners, key activities, key resources, value proposition, customer relationships, channels, customer segments, cost structure, and revenue streams. The business model canvas is quite widespread in the entrepreneurial context because it helps entrepreneurs to make sense of "doing business" to capture and deliver value to customers (Blank, 2013; Castillo-López et al., 2019; Massa & Tucci, 2013; Trimi & Berbegal-Mirabent, 2012). Thus, our effort was to enable all the participants to apply a forest cooperatives' integrated business model, develop a shared vision that is owned by all involved, for value creation and delivering, differentiation, introduction of innovative products and services and generation of economic and other resources. The business model canvas consists of nine main building blocks, forming and describing the integrated strategy that forest cooperatives may develop. These blocks are shaped in the following [Figure 1](#) (Osterwalder & Pigneur, 2010).

The overall process followed in this survey can be mapped as following:

a) Definition of the context of the Business Model Canvas and collection of data.

Key Partners	Key Activities	Value Propositions	Customer Relationships	Customer Segments
	Key Resources		Channels	
Cost Structure			Revenue Streams	

Figure 1. The business model canvas.

For that purpose, delegated discussions were made with key persons of the forest cooperatives in the region, such as presidents or members of the administrative councils of the cooperatives.

b) Identification of the core business model items and evaluation.

The respondents were asked to evaluate the significance of relevant items using a 5-point scale based on selected criteria. Scale anchors: 1 = “not significant”, 5 = “absolutely significant”.

c) Simplification of items and formulation of an integrated business model canvas.

This section includes the combination of respondents’ evaluation of several criteria into integrated meanings describing the building blocks of the proposed model, especially those that deal with key partners, activities and value propositions. This gave us the opportunity to extract activities, resources and value propositions that differ from the existing strategy of the forest cooperatives in the region, or they were not identified till now. We argue that this could give the opportunity for the cooperatives to focus their strategies in the proposed core elements regarding their operation in the future.

Thus, the first group of questions has to do with the identification of the key partners and their significance to the forest cooperative, the second with the key activities and the part they have today or will have in the future in the cooperative’s operation, the third with the main resources that are needed or are used, the fourth group with the value propositions especially under the new legislation context. The next group regarding customer relationships, channels and customer segments was stated as the marketing strategy of the business model, with the final group of questions for cost structure and revenue streams to follow. Finally, a group of questions referred to the profile of each cooperative. Likert scale was used for the majority of the questions. The items pertaining to each scale were pre-tested with face-to-face interviews that took place among representative cooperatives of the sample. This process allowed the assessment of the content validity of each question and ensure that surveyed cooperative representatives understood the content of the questionnaire (Dillman, 2000). The content validity of the questionnaires was also secured since some of the authors have significant experience on forest cooperatives in Greece, as members of the public forestry administration, which is responsible for the supervision of cooperatives. Furthermore, the business model canvas tool secures the validity of the group of questions that have been used.

Statistic analysis

We used the SPSS software for statistical data analysis (Version 24), and relative statistics tests were applied (Correlations, cross-tabulation, chi-square). Our analysis focuses mainly on value propositions as potential strategies that the forest cooperatives may develop, to adjust to the new legislation framework and the business environment. In order to simplify the generated canvas, we used factor analysis to determine which variables of the questionnaire we used, describe each building block of the canvas (Siomkos & Vasilikopoulou, 2005), especially those that deal with key partners, activities and value propositions. We used factor analysis according to the method of principal component analysis, in order to test the unidimensionality of the elements constituting each question, as well as the content validity of each question separately. Regarding the content validity of the research's variables, the statistical factor of Cronbach alpha was used (Wieland, Durach, Kembro, & Treiblmaier, 2017). Thus, a simplified business model canvas was generated. We argue that this was the first time that this grouping of each factor has been made regarding forest cooperatives in Greece and formulate a concrete business model of their operation in relation to the new legislation framework as well. We also argue that through this methodology an integrated strategy for forest cooperatives can be established, according to the determinants, opportunities, limitations and existing emerging parameters of the general business environment of the cooperatives.

Research questions

Based on the above-mentioned analysis and literature review, the paper constitutes a prototype research at the national level, aiming at answering the following research questions:

Q1: Can the business modelling approach be applied in forest cooperatives, using analytical tools such as the Business Model Canvas?

Q2: Which are the structural factors supporting the function of Greek forest cooperatives and the exploration of the possibilities of applying business model innovation to them?

Q3: How can the development of an integrated business model help forest cooperatives stay ahead in the product innovation game, while improving the sustainability of multi-functional forestry?

Q4: Which are the value propositions as potential strategies that the forest cooperatives may develop, in order to adjust to the new legislation framework and the business environment?

Q5: Can the business model approach applied to forest cooperatives develop a shared vision that is owned by all involved, for value creation and delivering, differentiation, introduction of innovative products and services and generation of economic and other resources?

Results

The business model canvas for forest cooperatives

In Figure 2, the analytical business model canvas is given for the forest cooperatives of the study, as it has been produced, taking under consideration the results regarding the significance of the specified factors which describe the building blocks of the canvas. From a significant number of factors, the most important are presented as they have been classified by the respondents, measuring the means of all the answers, with 1 as not significant at all and 5 as the most significant factor. Here, it should be noted that, in our results presentation, the most significant factors of each building block are shown, since we have measured a considerable number of variables using questionnaires.

Regarding their main resources, the interviewees believe that the new legislation framework does not give them the ability to differentiate from the existing situation, by stating that they intend to occupy certified forest workers (Mean = 4.94), as well as to be staffed with specialized personnel such as economists and accountants and other specialties (Mean = 4.51). Although this result can be considered as a positive advancement for their operation, thus far most of the cooperatives are operating as small family schemes that are not able to occupy specialized and scientific personnel, which could give them a competitive advantage and basic capabilities to operate in the new environment. Further, regarding human resources, there is a differentiation in the ranking of preferences, given that there is observed a higher rating in forestry specialty (Mean = 3.55) and specialty of foresters with technological education (Mean = 3.42).

As it is presented in the business model canvas (Figure 2), Greek forest cooperatives mainly depend financially on cooperatives' income and the collaboration with traders. Other

KEY PARTNERS	ACTIVITIES		VALUE PROPOSITIONS	MARKETING STRATEGY CUSTOMER RELATIONSHIPS CHANNELS SEGMENTS	
	- Timber felling and transport 4.86 - Environmental protection activities 4.12 - Commercial activities 3.72 - Tourist activities 3.05 - Wood processing and production of wood products 2.98			- Continuous communication with traders and other entrepreneurs 4.57 - Better pricing policy to regular and old customers 4.22 - Better pricing policy to new customers 3.95 - Establishment of an information mechanism from the customers 3.77 - Development of website 3.77 - Continuous support of customers regarding their emerging needs 3.75 - Awareness and information activities 3.72 - Sales of products according to the special characteristics of each customer 3.45 - Development of web solutions for supporting sales 3.37	
	KEY RESOURCES				
- Forest service 4.98 - Local administration 4.48 - Regional and central Administration 4.42 - Wood traders 4.35 - Other forest cooperatives 4.28 - Local communities and forest based population 4.09 - Equipment suppliers 4.02 - Wood processing industries 3.91	- Certified forest workers 4.94 - Economists - accountants 4.51 - Non certified forest workers 3.82 - Foresters 3.55 - Foresters with Technological Education 3.42 - Technicians 3.22 - Health and safety technical advisors 3.12 Equipment: Felling equipment, shifting animals, vehicles, other infrastructure - Comparative's income 4.92 - Collaborating traders 3.44 - Cooperative's reserves 3.14	- Development of knowledge basis and skills of the members 4.28 - Sustainable forest management 4.28 - Expansion of market for the cooperative 4.02 - Establishment of synergies with other organizations 3.94 - Enhancement of competitiveness 3.83 - Addressingmarketneeds 3.68			
COST STRUCTURE			REVENUE STREAMS		
- Production of timber products cost 4.29 - Personnel cost 3.97 - Cost of fixed and other equipment 3.84 - Administrative costs 3.15 - Invested capital cost (fixed and current assets) 3.04 - Cost of raw material and energy 2.84 - Taxation 2.76 - Credit costs 2.18			- Sales of firewood 4.54 - Sales of technical wood 4.26 - Sales of other roundwood 2.65 - Sales of industrial roundwood 2.02		

Figure 2. The analytical-proposed business model canvas for forest cooperatives (based on classification of means of each factor).

resources that have been examined, such as the funding projects and financial tools, the sales of innovative products or revenues that source from personnel training and knowledge, even the contribution of members, seem not to play a significant role. The value of the cooperative share constitutes a very basic feature of cooperatives. It is noted that the value of the cooperative share is mainly determined by the amount of funds the cooperative needs in order to finance its activities, with 55.4%. Then, with a percentage of 36.9, appears the criterion of the minimum amount set by the institutional framework of forestry cooperatives, while the criterion which is associated with candidate membership's income makes up a mere 7.7%.

Interesting conclusions also arise from the correlation of the criterion determining the value of the cooperative share and the number of active members of the cooperatives (Table 1). As it arises, the income of the candidate members is important only for the small cooperatives (less than 20 members). Almost respective are the findings related to the criterion of the minimum amount from the institutional framework. In particular, only 3 out of the 24 cooperatives who have declared this choice have more than 20 active members and none more than 24 (Table 1). Finally, it is noteworthy that the option, where the amount of the cooperative share is determined by the amount of funds needed to meet the financial requirements of the cooperatives, has a representation even in the very large cooperatives (it records a 6.2%).

Table 1. Cross tabulation of cooperative share amount and active members.

		Number of active members of cooperatives						Total
		<5	6–10	11–15	16–20	21–25	>25	
Criteria for determining the amount of the cooperative share	Income of candidate members	N 1	1	2	1	0	0	5
		% 1.5%	1.5%	3.1%	1.5%	0.0%	0.0%	7.7%
	Amount necessary to fund	N 2	11	10	5	4	4	36
		% 3.1%	16.9%	15.4%	7.7%	6.2%	6.2%	55.4%
	Minimum amount from the institutional framework	N 3	5	6	7	3	0	24
		% 4.6%	7.7%	9.2%	10.8%	4.6%	0.0%	36.9%
Total		N 6	17	18	13	7	4	65
		% 9.2%	26.2%	27.7%	20.0%	10.8%	6.2%	100.0%

Table 2. Factor analysis of forest cooperatives key partners.

Cooperation with:	Rotated Component Matrix ^a				
	Component				
	1	2	3	4	5
Universities/Technological Educational Institute	,768	-,002	,292	-,090	,108
Central government and regional authorities	,589	,306	-,135	,100	,373
Renewable energy companies and organizations	,577	,383	,295	,091	-,067
Non-governmental organizations (NGOs)	,557	,031	-,007	,081	-,084
Agricultural and other cooperatives	,125	,904	-,059	,026	,007
Other forest cooperatives	-,245	,748	,350	,112	,188
Suppliers of mechanical and other equipment	,327	,546	,100	-,095	,197
Industries of sawn wood	,002	-,017	,882	,172	-,169
Timber product industries	,153	,150	,794	-,037	,165
Local communities	,343	,338	,521	-,068	,239
Timber dealers	-,172	,053	,269	,846	-,160
Local authorities	,218	,051	-,152	,746	,313
Financial institutions	,526	-,072	,005	,559	,119
Local forest Services	,095	,246	,171	,164	,780
Businesses and traders of other non-wood forest products (NWFPs)	,390	,510	,203	,106	-,535

Extraction Method: Principal Component Analysis.

Rotation Method: Varimax with Kaiser Normalization.

^aRotation converged in eight iterations.

The above-mentioned lack of business financial strategy is also evident in the next two building blocks of the canvas, revenue streams, and cost structure, respectively. Results reveal, here as well, that the main revenue streams of the Greek forest cooperatives constitute only the sales of timber products (Figure 2). More specifically, these products are firewood (Mean = 4.54), with technical wood for constructions to follow (Mean = 4.26); thus, Greek forest cooperatives sell timber products of relatively low value. The cooperatives of the study state that until now, and we argue that this is their belief for the future under the new legislation context, they have had no or extremely limited revenues from products or process innovations adoption, from synergies with universities or research centers, from staff training and from investments they made and funding opportunities. Cost structure on the other hand, as it was expected, is mainly formed by the main cooperatives' activities such as the production of timber products (Mean = 4.29), personnel costs (Mean = 3.97), equipment costs (Mean = 3.84) and administrative costs (Mean = 3.15), with the other cost categories to follow as they are presented in the canvas (Figure 2). From this cost structure, we see that variable costs play the most significant part of the total operation cost for the Greek forest cooperatives. Hence, we argue that Greek forest cooperatives should move towards strategies for this cost reduction.

Finally, we examine the value propositions and the marketing strategy. These two building blocks, as we see above (Figure 2), in our opinion constitute the core elements for the Greek forest cooperatives, to develop an integrated business model and strategy regarding their operation in the context of the new legislation framework, as well as in the competitive business environment that they operate. Finally, the rest of the canvas building blocks is translated into a coherent value proposition of the cooperatives and ways to create, capture and deliver this value to customers. Hence, under the new legislation context, in particular, Greek forest cooperatives place knowledge and skills development at the core of their strategy (Mean = 4.28) for their development, operation and the creation and deliverance of value (Figure 2). The same significance holds the sustainable forest management as a value proposition. Furthermore, this value proposition, along with marketing strategies that the new legislation framework provides, is focused on continuous communication with traders and other entrepreneurs (4.57), expansion of market (4.02), creation of synergies with other organizations (3.94), enhancement of competitiveness (3.83), addressing market needs (3.68) along with better pricing policies (Figure 2).

Strategy of forest cooperatives

Alongside the exploration of the prospects that arise for the cooperatives from the implementation of the new law, how the current changes can shape their business model, and their strategy development is also investigated. Given that the recorded potential strategies interact with each other, an analysis based on the Pearson correlation coefficient (Pcc) was performed. As shown by the results, the strategies that positively influence each other with significance at 0.01 probability level ($p < .01$) are the following:

- Strategy of sustainable forest management correlated to the strategy of meeting market needs (Pcc = 0.427).
- Strategy of diversification and innovation correlated to the strategy of developing a business culture (Pcc = 0.436).

- Strategy to meet market needs correlated to the strategy of enhancing competitiveness (Pcc = 0.501).
- Strategy of market share expansion correlated to the strategy of enhancing competitiveness (Pcc = 0.646).
- Strategy of development of entrepreneurial culture, correlated to the market share expansion strategy (Pcc = 0.630).
- Strategy for the development of members' skills and knowledge, correlated to the strategy of forest business support (Pcc = 0.513).
- Strategy of support for forest enterprises/industries, correlated to the strategy of developing synergies with other cooperatives, etc (Pcc = 0.524).
- Strategy of developing synergies with Universities, correlated to the strategic development of entrepreneurial culture (Pcc = 0.443).
- Strategy for developing synergies with other cooperatives, correlated to the strategy of enhancing competitiveness (Pcc = 0.480).
- Strategy focuses on investments correlated to the strategy of developing synergies with Universities (Pcc = 0.412).

The above-mentioned results indicated that there exist quite strong correlations among the potential actions for adoption by forest cooperatives.

Structuring the integrated business model canvas for forest cooperatives

Factor analysis was applied to reduce the variables used to describe the building blocks of the business model canvas, using the existing correlations between these variables as a criterion. Our purpose is to propose the basic axis of each building block of the canvas that could stand as the main strategic orientation for forest cooperatives. Our analysis focuses on key partners, key activities, financial resources and of course value proposition and marketing strategy, since the rest of the factors that constitute the building blocks of the canvas, as they are presented at [Figure 2](#), constitute more or less fixed categories that are explained by the variables we used in our questionnaire and are not affected by the new legislation framework.

Thus, regarding key partners, the main factors that emerged (loadings min = 0.3) (Siomkos & Vasilikopoulou, 2005; Wieland et al., 2017) are presented in the next table. Hence, the main key partners of the Greek forest cooperatives could be grouped as (a) other research organizations, enterprises, and central administration, (b) other cooperatives and suppliers, (c) wood industry and forest population, (d) traders and financial organizations (e) forest service.

The content validity analysis ([Table 2](#)) regarding key partners of forest cooperatives has shown that the determinants (answers) that have been detected constitute concrete and reliable structures, able to contribute to the measurement of the factor to which they belong since they are close or over 0.7 (Siomkos & Vasilikopoulou, 2005). Thus, for the five factors describing key partners Cronbach's alpha = 0.634, 0.690, 0.712, 0.606, respectively. For key activities ([Table 3](#)), the main describing variables are (a) production of energy and non-timber products and services (b) wood processing and trading (c) lodging activities. Content analysis of each factor resulted Cronbach's alpha = 0.853, 0.667 and 0.410, respectively. The low value for Cronbach's alpha statistics ($\alpha = 0.410$) for lodging

Table 3. Factor analysis of forest cooperatives key activities.

Rotated Component Matrix ^a			
Key activities	Component		
	1	2	3
Cultivation and exploitation of energy plants, aromatic plants, etc	,833	,273	,152
Collection, processing, standardization and marketing of non-wood products	,793	,168	,056
Construction traditional and other wood products	,756	,201	,212
Production of renewable energy	,741	,163	,020
Forest tourism, recreation and mountain sports development	,565	,184	,521
Timber processing and production of wood products from your cooperation	,209	,801	,018
Commercial activities from your cooperative	,155	,735	,287
Charcoal, pellets etc	,477	,568	,041
Future logging operations	-,110	,219	,794
Environmental protection actions	,370	-,026	,765

Extraction Method: Principal Component Analysis.

Rotation Method: Varimax with Kaiser Normalization.

^aRotation converged in five iterations.

activities is interpreted by the fact that environmental protection activity is stated to be exerted in a moderate degree, showing a Mean value of 2.54.

Regarding key resources, based on the previous results (see Figure 2) we argue that for human resources, forest cooperatives under the new legislation framework should be staffed with the following categories of personnel: (a) Foresters and forest workers (b) administrative staff, economics, and accountants, (c) technicians and supporting staff. Material resources categories are also presented in Figure 2. Factor analysis (Table 4) for financial resources revealed that funding of forest cooperatives could derive from (a) External sources (b) Internal sources and member contributions (c) Sales of products and services (Cronbach's $\alpha = 0.769$, 0.594 for the two first categories).

Of great importance are the results of the next Table 5, describing the main value propositions categories and strategies deriving from them for the Greek forest cooperatives, under the new context, and in a competitive business environment. Factor analysis reveals the following three main value propositions that the cooperatives could adopt and develop (Table 5):

- Developing core capabilities for innovation and entrepreneurial activities
- Adopting an open innovation model
- Differentiation through sustainability

Table 4. Factor analysis of forest cooperatives financial resources.

Rotated Component Matrix ^a			
Financial resources	Component		
	1	2	3
Sponsorships	,904	-,090	-,045
Lending	,789	,096	,149
Funding programs	,720	,257	-,008
Extraordinary contribution of the members	,596	,585	,055
Cooperative's reserve	,032	,912	-,039
Revenue work carried out	,107	,261	-,732
Cooperating dealers	,197	,254	,721

Extraction Method: Principal Component Analysis.

Rotation Method: Varimax with Kaiser Normalization.

^aRotation converged in four iterations.

Table 5. Factor analysis for value propositions and strategy of forest cooperatives.

Strategies	Rotated Component Matrix ^a		
	Component		
	1	2	3
Developing members' skills and knowledge	,840	,035	-,066
Boost competitiveness	,723	,305	,139
Support forest-based enterprises/industries	,695	,233	,023
Meet market needs	,601	,149	,546
Developing synergies with other cooperatives etc.	,597	,259	,370
Development of entrepreneurial culture	,164	,761	,238
Focusing on investments	,105	,749	,082
Developing collaborations with Universities	,335	,642	-,071
Expanding the current market share	,456	,568	,138
Sustainable forest management	,077	-,017	,890
Diversification and innovation	-,013	,492	,622

Extraction Method: Principal Component Analysis.

Rotation Method: Varimax with Kaiser Normalization.

^aRotation converged in five iterations.

The content validity of the above factors resulted as Cronbach's $\alpha = 0.805$, 0.740 and 0.551 , respectively.

Finally, factor analysis for marketing strategy (customer's relationships, segments, and channels) has shown that all the factors we used in our analysis constitute basic variables in forming this strategy. This is due to the factors that we included in our research, which are based on the 4P's approach (product, price, place, and promotion), as elements for formulating a marketing strategy for a company or an organization (Constantinides, 2006; Kotler & Armstrong, 2010). The study of proposing an integrated marketing strategy and the specialization of this strategy in concrete measures and proposals could be a part of future research.

In Figure 3, the final form of Greek forest cooperatives' business model canvas is presented, as formed based on our previous analysis.

Discussion

The new legislation framework for forest cooperatives in Greece has set a new environment regarding their operation. Although it gives the opportunity for established and new ventures to evolve and incorporate contemporary business and entrepreneurial strategies to meet their role in a sustainable and multifunctional forestry approach, by developing innovations and applying capture and creation of value techniques, a significant percentage, which stands for the majority, is not ready for this transition and transformation. This result is however in line with relative research (Kajanus et al., 2019), performed in different contexts around Europe. A basic requirement for this shift, as we argue, is the formation of an integrated strategy by using the proper analytical tools for this effort, which can be applied for forest cooperatives too, regardless of the fact that they belong to a low-tech sector. Our analysis answers the first research question (Q1), by applying the BMC approach in the Greek forest cooperatives, constituting a prototype research at the national level and one of the few related researches internationally on the issue.

The results have highlighted that the cooperation with the Greek forestry service and the timber traders plays the most significant role for forest cooperatives and will do so in the future as well. This is due to the nature of forestry in general for Greece, since the majority of Greek forests are public. Here, we have to mention that, while forest

KEY PARTNERS a. Research organizations, enterprises and central administration b. Other cooperatives and suppliers c. Wood industry and forest population d. Traders and financial organizations e. Forest service	KEY ACTIVITIES a. Production of energy and non-timber products and services b. wood processing and trading c. Lodging activities.	VALUE PROPOSITIONS a. Developing core capabilities for innovation and entrepreneurial activities b. Adopting an open innovation model c. Differentiation through sustainability	MARKETING STRATEGY CUSTOMER RELATIONSHIPS CHANNELS SEGMENTS 4P's of forest cooperatives
	KEY RESOURCES Human resources a. Foresters and forest workers b. Administrative staff, several specialties c. Technicians and supporting staff. Equipment: Felling equipment, shifting animals, vehicles, other infrastructure Financial resources a. External sources b. internal sources and member contributions c. sales of products and services		
COST STRUCTURE Reduction of variable costs		REVENUE STREAMS a. Sales b. Other activities c. Innovative products and services	

Figure 3. Proposed Greek forest cooperatives' business model canvas.

cooperatives develop collaboration and synergies with other cooperatives or wood industries, this is not translated to the development of new products, innovations or the support of wood industries, which could be part of basic strategy proposals. Furthermore, the key activities of Greek forest cooperatives have to do mainly with timber felling and transport and environmental protection activities, such as the cultivation of forests. Forest cooperatives develop collaboration and synergies with other cooperatives or wood industries.

The above-mentioned are some of the first findings regarding the Q2 research question dealing with the structural factors supporting the function of Greek forest cooperatives and the exploration of the possibilities of applying business model innovation to them. Factor analysis has revealed these group of elements and characteristics of forest cooperative's operation structuring an integrated business model for forest cooperatives, constituting our research proposal with significant managerial and practical implications on the issue. The group of elements for this business model is dealing with development of cooperation and synergies with other cooperatives, research institutes and organizations, adoption of open innovation business models and differentiation through sustainability, recruitment of specialized personnel and effective human and other resources management, development of a concrete marketing strategy and finally the incorporation of innovative products, services and processes to their operating portfolio.

Regarding our Q3 research question concerning the ways that the development of an integrated business model can help forest cooperatives stay ahead in the product innovation game while improving the quality of Greek forests and environmental protection, our analysis of results shows that the new legislation framework does not give them the ability to differentiate from the existing situation. However, several results can be considered as a positive advancement for cooperatives' operation, since to date most of the cooperatives are operating as small family schemes that are not able to occupy specialized and scientific personnel, which could give them a competitive advantage and basic capabilities to operate in the new environment, a fact that is recognized as a prerequisite for their development (D'Amato et al., 2018; Kajanus et al., 2019; Todeschini et al., 2017; Trigkas et al., 2012).

Our results support the proposal for forest cooperatives to establish a framework for adopting and implementing activities related to sustainable forest management based on

cooperation and synergies creation bonds, for achieving a win-win strategy. Thus, they will be able also to satisfy the increasing needs for new and innovative forest-based products. This demand offers the opportunity for differentiation of forest products' markets but also constitutes a fundamental challenge for the sustainable management of forests and the balancing of demand (D'Amato et al., 2018). The demand for new forest-based products under the bioeconomy and bioenergy context should be coordinated to traditional practices without overcoming the sustainability boundaries.

Forest cooperatives are the basic organization for the implementation of sustainable forest management in Greece, under the guidance of public Forest Service. Our results show that, all the stakeholders, including forest cooperatives, should promote a holistic approach in forest management under its multifunctional context and take under consideration the whole value chain of forestry, determining the basic strategies that are required for the enhancement of sustainable forest management and the improvement of competitiveness, especially in rural areas. The business model approach gives a valuable tool for policymakers and forest cooperatives' administration to answer to this dilemma.

Greek forest cooperatives sell timber products of relatively low value. This is due to the economic crisis and recession Greece has been dealing with over the last 8 years and has resulted in the impressive rise of heating energy cost, mainly petrol, electricity and natural gas; firewood has been one of the most prevailing alternative energy sources for heating in Greece during the recession period. The cooperatives of the study state that until now, and we argue that this is their belief for the future under the new legislation context, they have had no or extremely limited revenues from products or process innovations adoption, from synergies with universities or research centers, from staff training and from investments they made and funding opportunities. However, the above factors are evaluated as crucial for their operation in producing and delivering innovations while improving the quality of Greek forests and environmental protection.

One of the findings with significant importance to our research is the value propositions as potential strategies that the forest cooperatives may develop, in order to adjust to the new legislation framework and the business environment (Q4), by applying the business modeling approach. The results of the canvas show that the Greek forest cooperatives seem to pay attention to this value creation and deliverance strategy meaning. However, this was not something they have mapped and planned until now, at least in an organized way. Value proposals of other business sectors and marketing strategies seem to play a fundamental role here, too. This is an ascertainment that is pointing to a positive direction regarding Greek forest cooperatives, although they belong to a low tech, traditional sector with limited innovation and differentiation capabilities. As we see, our business model canvas provides the opportunity for cooperatives to close the gap between knowledge about the effects of individual variables on their operation, and the need for knowledge on how relationships among the other variables of the canvas building blocks affect their operation and performance (Cosenz & Noto, 2018).

Greek forest cooperatives place knowledge and skills development at the core of their strategy, under the new legislation context as well, for their development, operation and the creation and deliverance of value. This is in alignment with the related findings in the issue of value creation and the role of knowledge basis, in other industries and sectors as well as value creation networks (Adner & Kapoor, 2010; Cambra-Fierro, Florin, Perez, & Whitelock, 2011; Johannessen & Olsen, 2010; Kim & Mauborgne, 1999; Löwendahl,

Revang, & Fosstenløyken, 2001; Möller & Svahn, 2006). Forest cooperatives remain the main business scheme in the primary forest sector in Greece that deals with timber and other forest products production, which ensures the application of sustainable forest management principles in forest exploitation, a value with strong economic, social and environmental impact.

Multiple correlations among independent strategies also exist, pointing to a much-required multidimensional approach for the forest cooperative's integrated strategy development. Major correlations deal with the strategy of sustainable forest management, correlated to the strategy of meeting market needs, the strategy of diversification and innovation, correlated to the strategy of developing a business culture, the strategy to meet market needs, correlated to the strategy of enhancing competitiveness and the strategy of support for forest enterprises/industries, correlated to the strategy of developing synergies with other cooperatives, etc. Business modeling approach has highlighted this complex strategy net into the organization's operation for the first time in the specific sector.

This is in line with the results describing the main categories of the value proposition of the forest cooperatives. We argue that strategy building and the creation of core capabilities and measures, which should be developed in order to meet the specified strategy, will constitute the main value propositions of forest cooperatives to operate in this competitive environment and survive, meeting their goals and mission of forest management and utilization to produce products and services of added value that meet several customer and market categories needs. The main value propositions that the cooperatives could adopt and develop are:

- Developing core capabilities for innovation and entrepreneurial activities
- Adopting an open innovation model
- Differentiation through sustainability

Finally, for our final research question (Q5), we examined the potential of the business model approach applied to forest cooperatives, to develop into a common vision shared by all parties involved, for value creation and delivering, differentiation, introduction of innovative products and services and generation of economic and other resources. The three above-mentioned value propositions could stand as the basis to this common vision for value creation in the specific sector, under the context of multifunctional and sustainable forest management. The business model approach defines the priorities and aims for action undertaking, connects strategies, boosts the coherence in planning, funding and implementing cross-sectoral activities throughout the whole forest value chain.

Conclusions

As stated in the beginning of the paper, forest cooperatives, like all joint venture, are essentially a form of entrepreneurship with emerging needs, such as economic development, diversification and enhancement of competitiveness, whose fulfillment requires the adoption of innovations as well as the introduction of new products, services and knowledge, aiming at creating value and meeting the modern citizens' needs. Thus, an integrated strategy is required to be developed for forest cooperatives too, in order to give them the opportunity to survive in the existing competitive business environment, by providing

value propositions through innovations, contemporary marketing strategies, creation of synergies, cost-effective operation and meeting of their customer and members' needs, while ensuring sustainable forests management under their multifunctional context. Business models work as a tool capable to bridge the gap between knowledge about the nature, the relations and the way of individual variables on firm performance and value creation. The structural elements of a business strategy must be concrete and well tempered, so between them as well as between a firm's strategy and its business model.

We argue that the present research contributes to the strategy theory of the forest sector and more specifically for forest cooperatives and related forms of entrepreneurship, by extending research in the field (i.e., of Atmiş et al., 2009; García-López, 2018; Kajanus et al., 2019; Szulecka et al., 2016; Vega & Keenan, 2016) and providing a value creation-oriented approach, of distinguished practical implications as well. As stated, our findings are in line with related research in other industrial and business context (Adner & Kapoor, 2010; Cambra-Fierro et al., 2011; Cosenz & Noto, 2018; Johannessen & Olsen, 2010; Teece, 2018) and highlights the fact that business modeling and value creation through innovations, differentiation and building of capabilities in specific business operations (Chesbrough & Rosenbloom, 2002; Osterwalder & Pigneur, 2010; Teece, 2018), can also be applied in the forest cooperative's sector, regardless the fact that is a low-tech sector (D'Amato et al., 2018; Kajanus et al., 2019).

The application of the Business Model Canvas approach gives an opportunity for forest cooperatives to understand the new business environment that emerges and implement concrete measures for developing well-shaped strategies for value creation, growth and differentiation and enhancement of sustainability in the forestry sector. We argue that this was the first time that this grouping of each factor has been made regarding forest cooperatives in Greece and have formulated a concrete business model for their operation, also in relation to the new legislation framework, based on inputs from all of the surveyed cooperatives. We also argue that through this methodology an integrated strategy for forest cooperatives can be established, according to the determinants, opportunities, limitations and existing emerging parameters of the general business environment of the cooperatives.

Limitations

Of course, the present paper presents specific limitations which could stand as sources for future research in the field. First, our research is limited to a specific region of Greece and the cooperatives operating in this region, the region of Thessaly. The research could be further applied to the whole country by surveying the sum of the Greek forest cooperatives in order to extract more representative results and findings. Furthermore, the present research could act as a feedback survey in order to focus in a more thoughtful future research regarding the structural elements of the building blocks of the Business Model Canvas for forest cooperatives, by proposing strategies, value propositions and key stakeholders of principal meaning for the cooperatives. Finally, we choose Osterwalder's and Pigneur's Business Model Canvas approach as one of the most frequent methodologies for establishing a business model. Our research could be further approached by applying several other businesses' model design and techniques to highlight differences and similarities in findings, among the different approaches, especially for the forest cooperatives and the forestry sector.

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